Calculus and Linear Algebra

Monotonicity and Extrema

Lecture 21

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Lecture outline

- From the Mean Value Theorem to monotonicity.
- Increasing and decreasing intervals from the sign of f'.
- Critical points and first-derivative sign charts.
- Local maxima and minima: first derivative test.
- Absolute extrema on closed intervals.
- Optimization problems and economic interpretation.

Main goal

Use f' not only to compute slopes, but to reconstruct the qualitative behaviour of f.

Recall: the Mean Value Theorem

If f is continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Interpretation

Some instantaneous slope equals the average slope over the interval.

This result turns information about f' into information about the whole function.

Derivative sign and monotonicity

Theorem Let f be differentiable on an interval I.

- If $f'(x) > 0$ for every $x \in I$, then f is strictly increasing on I.
- If $f'(x) < 0$ for every $x \in I$, then f is strictly decreasing on I.
- If $f'(x) = 0$ for every $x \in I$, then f is constant on I.

Key message

The sign of f' determines the direction of motion of the graph.

Why the theorem is true

Take $x_1 < x_2$ in I. By the Mean Value Theorem,

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$$

for some $c \in (x_1, x_2)$. If $f'(c) > 0$ and $x_2 - x_1 > 0$, then

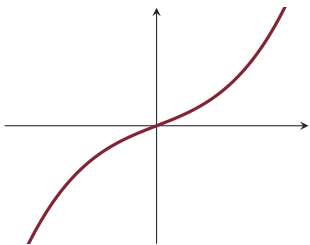
$$f(x_2) - f(x_1) > 0,$$

so

$$f(x_2) > f(x_1).$$

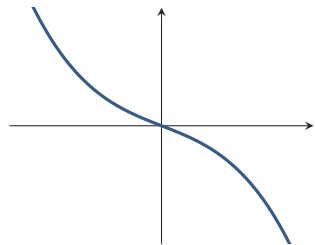
The decreasing case is analogous.

A visual dictionary



$f'(x) > 0$:

increasing



$f'(x) < 0$:

decreasing

Critical points: what must be checked?

A point x_0 in the domain is critical when

$$f'(x_0) = 0$$

or when $f'(x_0)$ does not exist. Critical points divide the domain into intervals on which the sign of f' can be studied.

Workflow

Domain $\to$ derivative $\to$ critical points $\to$ sign of f' $\to$ monotonicity.

First derivative test

Suppose x_0 is a critical point.

Sign of f' before x_0	Sign after x_0	Conclusion
$+$	$-$	local maximum
$-$	$+$	local minimum
$+$	$+$	no extremum
$-$	$-$	no extremum

Important

$f'(x_0) = 0$ alone is not enough. We must inspect the sign change.

Sign chart: the standard notation

Suppose

$$f'(x) = (x+1)(x-2).$$

The critical points are -1 and 2.



Hence $x = -1$ is a local maximum and $x = 2$ a local minimum.

Example 1: a cubic polynomial

Exercise Study monotonicity and local extrema of

$$f(x) = x^3 - 3x^2 + 2.$$

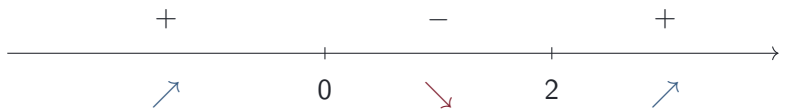
$$f'(x) = 3x^2 - 6x = 3x(x-2).$$

Critical points:

$$x = 0, \qquad x = 2.$$

We now study the sign of $3x(x-2)$.

Example 1: derivative sign chart



Therefore:

$$\text{increasing on } (-\infty, 0) \cup (2, +\infty),$$

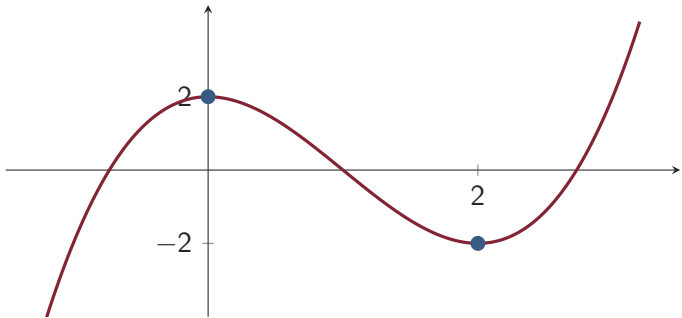
$$\text{decreasing on } (0, 2).$$

Moreover,

$$f(0) = 2, \qquad f(2) = -2.$$

So $(0, 2)$ is a local maximum and $(2, -2)$ a local minimum.

Example 1: graph confirms the analysis



The derivative sign chart predicts exactly the turning behaviour of the graph.

Example 2: a stationary point without extremum

Consider

$$f(x) = x^3.$$

Then

$$f'(x) = 3x^2 \geq 0, \qquad f'(0) = 0.$$

But f' is positive on both sides of 0.

$$\boxed{x = 0 \text{ is not a local maximum or minimum.}}$$

The function remains increasing through the stationary point.

Example 3: a rational function

Exercise Study monotonicity of

$$f(x) = \frac{x}{x^2+1}.$$

The domain is all of $\mathbb{R}$, and

$$f'(x) = \frac{(x^2+1) - 2x^2}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2}.$$

Since the denominator is always positive, the sign of f' is the sign of

$$1 - x^2 = (1-x)(1+x).$$

Example 3: classification

Critical points:

$$x = -1, \qquad x = 1.$$



Thus f decreases, then increases, then decreases.

$$f(-1) = -\frac{1}{2}, \qquad f(1) = \frac{1}{2}.$$

So $(-1, -1/2)$ is a local minimum and $(1, 1/2)$ a local maximum.

Absolute versus local extrema

A local extremum compares values only nearby. An absolute extremum compares with the whole domain:

$$f(x_M) \geq f(x) \quad \forall x \in D$$

for an absolute maximum, and similarly for a minimum.

Connection with Weierstrass

A continuous function on a closed bounded interval $[a, b]$ attains both an absolute maximum and an absolute minimum.

Closed-interval method

To find absolute extrema of a continuous function on $[a, b]$:

1. Find all critical points in (a, b).
2. Evaluate f at every critical point.
3. Evaluate $f(a)$ and $f(b)$.
4. Compare all values.

Why endpoints matter

Fermat concerns interior differentiable extrema. Absolute extrema may occur at endpoints.

Example 4: absolute extrema on $[-1, 3]$

Let

$$f(x) = x^3 - 3x^2 + 2.$$

Critical points inside the interval are 0 and 2.

x	-1	0	2	3
$f(x)$	-2	2	-2	2

Hence

$$\max_{[-1,3]} f = 2, \qquad \min_{[-1,3]} f = -2.$$

The absolute extrema are attained at more than one point.

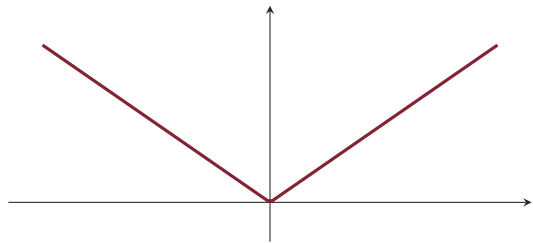
A derivative may fail to exist at an extremum

For

$$f(x) = |x|,$$

$x = 0$ is an absolute minimum, but

$$f'(0) \text{ does not exist.}$$



This is why nondifferentiable domain points must also be considered among critical points.

Optimization: translating words into mathematics

Many applications ask for the largest or smallest value of a quantity. Typical structure:

1. Identify the quantity to optimize.
2. Express it as a function of one variable.
3. Determine the admissible domain.
4. Find critical points.
5. Compare relevant values.
6. Interpret the answer in context.

Economic example: profit maximization

Suppose revenue and cost are

$$R(q) = 60q - q^2, \qquad C(q) = 20q + 100,$$

for $0 \leq q \leq 50$. Profit is

$$\Pi(q) = R(q) - C(q) = 40q - q^2 - 100.$$

Then

$$\Pi'(q) = 40 - 2q.$$

The interior critical point is

$$q = 20.$$

Economic example: checking the maximum

Since

$$\Pi'(q) > 0 \quad \text{for } q < 20, \qquad \Pi'(q) < 0 \quad \text{for } q > 20,$$

profit increases before 20 and decreases after 20. Therefore

$$\boxed{q = 20 \text{ maximizes profit.}}$$

$$\Pi(20) = 40(20) - 20^2 - 100 = 300.$$

Economic interpretation

The optimum occurs where marginal profit changes from positive to negative.

Marginal revenue and marginal cost

Because

$$\Pi(q) = R(q) - C(q),$$

we have

$$\Pi'(q) = R'(q) - C'(q) = MR(q) - MC(q).$$

At an interior differentiable optimum,

$$\Pi'(q) = 0 \quad \Longrightarrow \quad \boxed{MR(q) = MC(q).}$$

This condition identifies a candidate optimum; the sign change still determines whether it is a maximum.

Exercise: monotonicity

Exercise For

$$f(x) = x^3 - 12x,$$

find:

1. all critical points;
2. intervals of increase and decrease;
3. all local extrema.

Exercise: solution

Solution

$$f'(x) = 3x^2 - 12 = 3(x-2)(x+2).$$

Critical points: $x = -2, 2$.

$$f' > 0 \text{ on } (-\infty, -2) \cup (2, +\infty),$$

$$f' < 0 \text{ on } (-2, 2).$$

Therefore

$$(-2, 16) \text{ is a local maximum,} \qquad (2, -16) \text{ is a local minimum.}$$

Exercise: rational function

Exercise Study monotonicity and extrema of

$$f(x) = x + \frac{4}{x}, \qquad x \neq 0.$$

Hint:

$$f'(x) = 1 - \frac{4}{x^2} = \frac{x^2 - 4}{x^2}.$$

Remember that $x = 0$ is not in the domain and divides the analysis into separate intervals.

Exercise: solution

Solution Critical points are $x = -2, 2$; $x = 0$ is excluded from the domain.

$$f'(x) > 0 \text{ for } |x| > 2, \qquad f'(x) < 0 \text{ for } 0 < |x| < 2.$$

Hence

$$(-2, -4) \text{ is a local maximum}, \qquad (2, 4) \text{ is a local minimum}.$$

The domain separation at 0 is essential.

Common mistakes

- Solving $f'(x) = 0$ and stopping there.
- Forgetting points where f' does not exist but f does.
- Treating points outside the domain as critical points.
- Forgetting endpoints in absolute-extrema problems.
- Confusing a stationary point with an extremum.
- Ignoring disconnected parts of the domain.

Independent practice

Exercise Study monotonicity and local extrema of:

$$\text{A: } f(x) = x^4 - 4x^2,$$

$$\text{B: } g(x) = \frac{x^2}{x^2+1},$$

$$\text{C: } h(x) = \ln x - x, \qquad x > 0.$$

For each function, draw a sign chart for the derivative before sketching the graph.

Practice A: answer

For

$$f(x) = x^4 - 4x^2, \qquad f'(x) = 4x(x^2 - 2),$$

critical points are

$$-\sqrt{2}, \quad 0, \quad \sqrt{2}.$$

The derivative signs are

$$-,\ +,\ -,\ +.$$

Thus there are local minima at $x = \pm\sqrt{2}$ and a local maximum at $x = 0$.

Practice B and C: answers

For

$$g(x) = \frac{x^2}{x^2+1}, \qquad g'(x) = \frac{2x}{(x^2+1)^2},$$

g decreases on $(-\infty, 0)$ and increases on $(0, +\infty)$: $(0, 0)$ is a minimum. For

$$h(x) = \ln x - x, \qquad h'(x) = \frac{1}{x} - 1 = \frac{1-x}{x},$$

h increases on $(0, 1)$ and decreases on $(1, +\infty)$: $(1, -1)$ is a maximum.

What to remember

$$\boxed{f' > 0 \Rightarrow f \text{ increasing}} \qquad \boxed{f' < 0 \Rightarrow f \text{ decreasing}}$$

$$+ \to - \Rightarrow \text{local maximum}, \qquad - \to + \Rightarrow \text{local minimum}.$$

Next step

The first derivative describes direction. The second derivative will describe how the slope itself changes: concavity and convexity.